

REVEALING THE NATURE OF OVERLAPPING PATHS: ON PARASITIC GAPS, CYCLIC LINEARIZATION, AND MULTIPLE SPECIFIER FORMATION*

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1 Introduction

I examine structures where movement paths overlap. These minimally involve two moving phrases, whose paths will either cross, with no change in order (1a), or nest, reversing their order (1b):

- (1) a. $\alpha \beta \dots \alpha \beta$ (*Crossing paths*) b. $\beta \alpha \dots \alpha \beta$ (*Nesting paths*)
-

See Richards (1997) for cross-linguistic examples of both scenarios. In English, overt overlapping A'-movements always form nesting paths, as described by the Path Containment Condition (PCC; Pesetsky 1982). This is exemplified by (2) below. Here topic/focus fronting targets a peripheral position preceding a *wh*-moved phrase (Radford 2009, pg. 329), with obligatory nesting paths:

- (2) a. [This book]₂, who₁ should we talk to *t*₁ about *t*₂? [2 1 ... 1 2]
b. * [This student]₁, what₂ should we talk to *t*₁ about *t*₂? *[1 2 ... 1 2]

This paper's first goal is to examine the syntactic derivation involved in overlapping movement. Research in phase theory (Chomsky 2000, 2001) argues that a clause involves at least two phasal domains, the CP and the vP. Any movement through these phrases must pass through their edge. The fact that there are two moving phrases in the acceptable sentence (2a) above means that there are two conceivable structures for successive-cyclic movement through the vP here. In principle, the topic/focus phrase might form an outer specifier of vP, with the *wh*-phrase forming an inner one (3a), or the reverse (3b), after which they move on to their designated final landing sites:¹

*Contact: colind@alum.mit.edu. I am thankful for feedback from Kenyon Branan, Justin Colley, Patrick Elliott, Michael Yoshitaka Erlewine, Danny Fox, Sabine Iatridou, Laura Kalin, Martia Martinovič, David Pesetsky, Norvin Richards, Shoichi Takahashi, the audiences of NELS 50, WCCFL 38, MIT, the University of Konstanz, and to the people I compelled to judge the complex sentences investigated here.

¹These phrases may well ultimately land in the specifiers of separate projections within a multi-phrasal left periphery (Rizzi 1997), but this level of detail is not important here.

- (3) a. [_{CP} *this book*₂ *who*₁ ... Subj [_{vP} ***this book*₂ *who*₁** v-V ... *who*₁ *this book*₂]]
 b. [_{CP} *this book*₂ *who*₁ ... Subj [_{vP} ***who*₁ *this book*₂** v-V ... *who*₁ *this book*₂]]

I argue that facts about parasitic gaps (PGs; Engdahl 1983, a.o.) reveal that the derivation in (3a) is correct, when considered along with predictions from Nissenbaum (2000) about the interaction of PGs and multiple specifiers. In (3a), the A'-moved phrase with the higher surface position (*this book*₂) was also the higher specifier formed by successive-cyclic movement through the vP. I argue that this correspondence in intermediate and final orders is what we predict under the *Cyclic Linearization* approach to phases (CL; Fox and Pesetsky, 2005a,b, a.o.).

Further, this paper argues that CL also derives the PCC. Richards (1997, §2.6.2.2.1) hypothesizes that PCC effects correlate with the availability of only one overt specifier of CP in a given language, and that the PCC thus disappears in languages that permit multiple overt specifiers of CP. In this vein, I argue that the PCC emerges from CL given the proposal that in languages like English, A'-movement to lower specifiers is covert (Richards, 1997, Nissenbaum, 2000, a.o.). I show that this theory makes correct predictions about the PCC-lacking language Bulgarian. Along the way, I will argue that crossing and nesting specifier formation are both in principle permitted, unless a single triggering feature is involved, in which case crossing paths are required (building on McGinnis 1998, Rackowski 2002, Doggett 2004, Rackowski and Richards 2005, Kotek 2014).

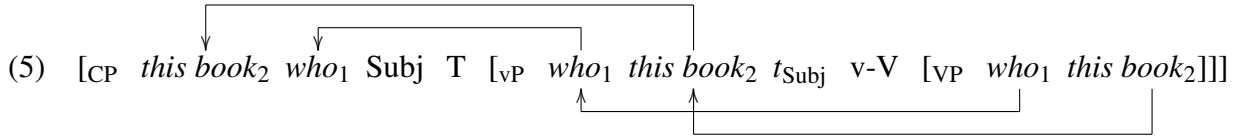
2 Background: Cyclic Linearization and multiple specifiers

Chomsky (2000, 2001) argues that phrasal movement must pass in successive-cyclic fashion through the edge (specifier) of phasal constituents, including at least vP and CP. The phase edge is an escape hatch for movement because it is not subject to phase-level spell-out, which targets only the phase's complement, and freezes any material that it applies to. This is known as the Phase Impenetrability Condition. Thus, in this theory, material moving out of the complement of a phase must reach the phase's specifier before the complement spells-out. In contrast, the CL theory initiated by Fox and Pesetsky (2005a,b) argues that spell-out applies to entire phasal phrases, including their head and specifier (see also Davis (2020a, 2021)). Thus CL must abandon the hypothesis that spell-out-avoidance motivates successive-cyclicity, since in this theory, nothing escapes spell-out. Instead, CL argues that successive-cyclic movement, which is indeed subject to spell-out in each phase, maintains the consistency of the linear orders established by spell-out throughout the derivation. I will argue that the CL theory makes accurate predictions about the structure of overlapping path configurations, which involve multiple specifier formation in vP.

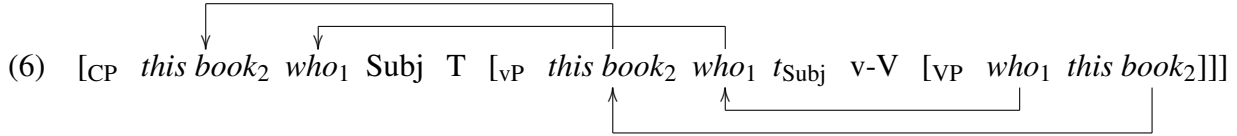
Richards (1997, 1999) and much work following argues that when two phrases move to specifiers of one head, the closest phrase is attracted first (4a). The lower one which is attracted second "tucks-in" to a lower specifier (4b), since this forms the shortest possible path:

- (4) a. [_{XP} α X ... [_{YP} ... t_α β]] b. [_{XP} α β X ... [_{YP} ... t_α t_β]]
-

Consider what this theory predicts for successive-cyclic movement through vP in the multiple A'-movement scenario we saw in (2a) above. First, the initially higher phrase *who*₁ should move to spec-vP. Second, the lower phrase *this book*₂ should tuck-in to a lower spec-vP below *who*₁. Next, these moving phrases will then continue on to their final landing sites (5):



This theory rules out a derivation in which, at the vP level, the lower phrase *this book*₂ moves above *who*₁ to form a higher specifier of vP (6), rather than tucking-in beneath it:



However, I argue that this non-tucking-in derivation, which CL predicts, is the correct one. My argument for this is based on facts about PGs, as explained in section 3 below.

After this, the analysis in section 4 will hypothesize that second specifier formation is free to tuck-in as in (5), or form a higher specifier as in (6), but that in English, the tucking-in derivation in (5) is filtered out for linearization reasons. This leaves (6) as the only option. In section 5, I refine this hypothesis with a more specific proposal about the triggering of tucking-in.

3 The relationship between parasitic gaps and specifier order

PGs are gaps in islands which co-refer with a phrase undergoing A'-movement structurally across that island (Engdahl 1983, Culicover and Postal 2001). PGs involve movement of an operator to the edge of the island (Chomsky 1986, Browning 1987, Nissenbaum 2000, a.o.). Nissenbaum (2000) argues that PGs are formed by the operator-hosting phrase adjoining to a landing site of successive-cyclic movement. His account of basic PGs in sentential adjuncts like that in (7) below is as follows. The operator moves to the edge of the island, triggering Predicate Abstraction (Heim and Kratzer 1998). If the island is a sentential adjunct as in (7), its original type *t* is thus raised to $\langle e, t \rangle$ (assuming semantic vacuity of the operator). Also, the licensing phrase that ultimately binds the PG successive-cyclically A'-moves through vP. This triggers Predicate Abstraction, creating an $\langle e, t \rangle$ position (*v'*₁) in it. The type $\langle e, t \rangle$ adjunct containing the PG can adjoin to that $\langle e, t \rangle$ node in vP, and combine with it by Predicate Modification (Heim and Kratzer 1998). This conjoins their denotations, creating another $\langle e, t \rangle$ node in vP (*v'*₂). This new $\langle e, t \rangle$ *v'*, marked with a box in (7), is saturated by the trace of successive-cyclic movement from vP. Assume that successive-cyclic movement in vP precedes merger of the adjunct, given that adjuncts merge late (Lebeaux 1988, a.o.). Late adjunction renders the intermediate trace of the licenser and its corresponding λ non-local, though still interpretable. Ultimately, the moved phrase binds its original trace in VP, and the operator's trace in the adjunct, which is the PG:

- (7) *The derivation of a PG:* Who_j did you [vP: *t*_j [$\boxed{v'2: \langle e, t \rangle}$ [*v'*_{1: $\langle e, t \rangle$} λ_j forget about *t*_j] [*Adjunct: $\langle e, t \rangle$* OP_k λ_k after talking to *t*_k(=PG)]]]?

Furthermore, when two phrases move through vP as in (8) below, Nissenbaum argues that recursive applications of Predicate Abstraction create an $\langle e, \langle e, t \rangle \rangle$ node (*v'*₁ below). This is saturated by the trace of whichever phrase formed the lower specifier of successive-cyclic movement through vP. That saturation yields an $\langle e, t \rangle$ node (*v'*₂ below), which is saturated by whichever phrase formed the higher trace of movement through vP, as (8) shows:

- (8) *Two movements*: $XP_j YP_k \dots [_{vP:t} t_j [_{v'2:<e,t>} t_k [_{v'1:<e,<e,t>}> \lambda_k \lambda_j v [_{VP} \dots t_j t_k]]]]$

We have seen in (7) that a PG-containing adjunct is type $\langle e,t \rangle$. In (8), in order for the PG to be licensed by the trace of the phrase that moved through the lower specifier in vP (t_k), it is necessary to merge the adjunct to the lower v' , which is $v'1$ in (8) above. Since $v'1$ is type $\langle e,\langle e,t \rangle \rangle$ due to being subjected to Predicate Abstraction twice, the PG-containing adjunct of type $\langle e,t \rangle$ cannot combine with it by Predicate Modification or Functional Application. The only option is for this adjunct to attach to the higher v' ($v'2$), which is type $\langle e,t \rangle$, and thus allows Predicate Modification with the adjunct to succeed. Importantly, merger to this position will result in the adjunct being saturated by the trace of the higher phrase that moved through vP . This reasoning leads Nissenbaum to the prediction that when multiple phrases move through vP , only the one that moved through a higher specifier of vP should be able to license a PG.

As described above, CL predicts that the higher moved phrase on the surface in an overlapping path derivation should have also been the higher specifier of successive-cyclic movement through vP . If Nissenbaum's reasoning about PGs is correct, the higher phrase within vP should also be the only available PG-licenser. I argue that this is correct. The higher phrase on the surface in such a configuration is also the only available PG-licenser, as (9) demonstrates:²

- (9) *A PG is licensed only by the higher of two moved phrases*
- a. [This book]₂, who₁ should we talk to t_1 about t_2
 ✓...before commenting on PG₂? / *...before arranging a meeting with PG₁?

While (9) is a mono-clausal configuration, in English we see overlapping A'-movements in multi-clausal constructions which contain multiple CPs that can serve as A'-landing sites. Such sentences must obey the PCC, as Pesetsky (1982) shows. Some examples of this form are provided in (10-12) below. Each of these also includes a sentential adjunct with a PG. As in (9) above, only the outermost of the two overtly A'-moved phrases licenses the PG:³

- (10) a. [**What car**]₂ did you realize **who**₁ to persuade t_1 to buy t_2
 ✓...in order to get to borrow **PG**₂ for free?
 *...after having a few drinks with **PG**₁?
- (11) a. [**Which movie**]₂ did you decide **who**₁ to ask t_1 to rent t_2
 ✓...in order to get a chance to watch **PG**₂?
 *...after giving **PG**₁ some money?
- (12) a. This is the book **which**₂ I asked **who**₁ you gave t_1 comments on t_2
 ✓...after making yourself a copy of **PG**₂
 *...after having a discussion with **PG**₁

²The judgments reported in (9-12) are corroborated by verbal surveys with eight native English speakers available in my local area, including five American, two Australian, and two British speakers. I acknowledge that one additional American speaker reported the reverse judgment, requiring PG-licensing by the inner moved phrase. There is a cross-linguistic parallel to this: Heck and Himmelreich (2017) show data from German, such as their (13a), where the lower of two moved phrases licenses a PG. This is a puzzle that should be addressed, but is beyond the scope of the present paper.

³den Dikken (1995) argues that indirect objects cannot undergo null operator movement. This might account for the failure of PG licensing in (11b), though this example still falls under the more general pattern described here.

Nissenbaum and Schwarz (2011) argue that parasitic gaps inside of subject islands are licensed by the same semantic mechanisms as those in adjuncts. Thus we expect them to show the same asymmetry. This prediction is verified by an earlier observation from Pesetsky (1982):

- (13) a. This volvo is one car **which**₂ I know **who**₁ to persuade [[owners of $\checkmark$ PG₂] / [friends of *PG₁] to talk to t_1 about t_2]. (Adapted from Pesetsky 1982, ch. 4, ex. 81-82)

In summary, given Nissenbaum's theory of PGs, the fact that only the higher phrase on the surface can license a PG in overlapping movement configurations indicates that this phrase was also the outer specifier of successive-cyclic movement from vP. This is as CL predicts.

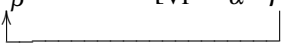
4 Deriving the Path Containment Condition

We have now seen evidence that the final relative order of two successive-cyclically moving phrases is matched in their intermediate landing sites in the vP edge. This achieves the coherence in phase-by-phase spell-out that CL predicts. However, nothing said so far explains why overlapping movement derivations in English must obey the PCC. I address this next.

Recall the proposal from Richards that multiple movements to one head result in tucking-in. Richards argues for tucking-in in part based on languages like Bulgarian, where all *wh*-phrases in a multiple *wh*-question overtly move to specifiers of CP, as in (14a). In contrast, for Richards and also Nissenbaum (2000), the difference between this and its English equivalent in (14b) is that for English, tucking-in of the second *wh*-phrase to a lower spec-CP involves covert movement:

- (14) a. *Bulgarian* (Richards, 1997, ex. 1) b. *English*
Kogo₁ **kavo**₂ e pital Ivan t_1 t_2 ? **Who**₁ $\emptyset$ ₂ did he ask t_1 **what**₂?
 whom what AUX asked Ivan

With this in mind, consider an overlapping overt movement configuration in English, such as (2a) above. If this involves tucking-in in vP, this would generate a vP like that schematized in (15) below. Here the phrases α and β both move to the vP edge, and β tucks-in below α . Given the proposal from Richards and Nissenbaum that tucked-in movement in English is covert, we should regard the tucking-in of β here to involve covert movement:

- (15) *Covert tucking-in*: [_{vP} α $\emptyset_\beta$ S v-V [_{vP} t_α β]] Linearization: $\alpha < S < V < \beta$
- 

Since α was linearized at the left edge of vP in (15), CL allows it to overtly move on. This is schematized in the hypothetical derivation for a CP in (16) below, where following (15), α and β both overtly move from vP. However, because β did not overtly reach the left edge of vP, its further overt movement would yield contradictory linearization information. Specifically, (15) demands that β follow the (rest of the) material originating in vP, while (16) requires that β precede that same material. Thus the linearizations of vP in (15) and CP in (16) conflict:

- (16) *Later overt movement of tucked-in β from vP: Contradicts (15)*
 [_{CP} α ... β ... [_{TP} S [_{vP} t_α t_β ... v-V ... t_α t_β]]] Linearization: $\alpha < \beta < S < V$

If this were a multiple-*wh* question where the movement of β beyond vP were covert, as in (14b) above, there would be no problem. This is because covert movement does not affect linearization.⁴ However, if β moved overtly from vP, as in any derivation where both moving phrases are overtly displaced (such as (2a) above), the result is an incoherent linearization. This problem is avoided if, instead of tucking-in below α in vP as in (15), β moves to a specifier above it within vP as in (17) below. Here neither β nor α covertly move, since neither tucks-in.⁵ From here, further overt movement of β and α beyond vP is licit, as long as they maintain their relative order:

(17) *Not tucking-in within vP yields two overt movements to spec-vP*

$$[_{vP} \beta \alpha S \text{ v-V } [_{VP} t_\alpha t_\beta]]$$
Linearization: $\beta < \alpha < S < V$

In summary, in order to avoid a linearization issue caused by covert tucking-in in English, these moving phrases reverse their order. This derives the PCC. I hypothesize that crossing and nesting multiple specifier formation are both permitted, but that in PCC configurations, crossing paths in vP are ruled out at PF due to a linearization conflict, leaving nested paths as the only option.

4.1 The possibility of overt inner specifiers voids the PCC

I have argued that crossing and nesting multiple specifiers are both syntactically licit, but that in English, the covertness of tucked-in inner specifiers sometimes interacts with linearization to force order reversal, and hence the PCC. This predicts that in a language which allows overt inner specifiers, the PCC should not arise. Bulgarian is a language that permits overt inner specifiers (Rudin 1988, Richards 1997) as we saw in (14a). Indeed, Rudin (1988) shows Bulgarian examples with overlapping moved phrases that demonstrate the expected freedom of ordering:

- (18) a. (Crossing paths, adapted from Rudin 1988, ex. 8c)
 Čoveka [**koj**_{to1} ne znaeš [**kakvo**₂ kazvat [če e kupil *t*₁ *t*₂]]]
 man who.REL NEG know.2S what say.3P that has bought
 ‘The man who you don’t know what they say that he bought’
- b. (Nesting paths, adapted from Rudin 1988, ex. 19)
 Edna kniga [**kojato**₁ se čudja [**koj**₂ *t*₂ znae [**koj**₃ *t*₃ prodava *t*₁]]]
 a book which wonder.1S who knows who.REL sells
 ‘A book which I wonder who knows who sells (it)’

Example (18b) involves two *wh*-movements nested within relativization. While just two paths would be sufficient to make the point, the presence of the third does not affect the argumentation.

5 Tucking-in is required only when a single feature attracts

Though Bulgarian allows either crossing or nesting in many cases, this and various other multiple *wh*-fronting languages require crossing paths in multiple-*wh* questions with two moving phrases:

⁴In fact, the behavior of β would be the same if it simply did not move in vP at all. CL would permit covert movement of β directly to spec-CP later on, since there is no independent reason for it to move successive-cyclically in this case. Nissenbaum (2000) argues that the lower *wh*-phrase in English multiple-*wh* questions does covertly move successive-cyclically through vP in at least some configurations, however.

⁵I hypothesize that α is not interpreted by PF as covertly moving here, despite the fact that it is ultimately an inner specifier, because it did not tuck-in.

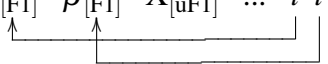
(19) *Order of Bulgarian multiple wh-movement* (Richards 1997, p. 277, ex. 75)

- a. Kogo₁ kakvo₂ e pital Ivan t₁ t₂? b. *Kakvo₂ kogo₁ e pital Ivan t₁ t₂?
 who what AUX asked Ivan what who AUX asked Ivan
 ‘Who did Ivan ask what?’ ‘What did Ivan ask who?’

The same is true in English, which does not permit anything resembling nesting paths in normal multiple-*wh* questions. This is demonstrated in (20), which continues to assume that any tucking-in movements in English would be covert:

- (20) a. ✓ **Who**₁ ∅₂ did Ivan ask t₁ **what**₂?
 b. * **What**₂ ∅₁ did Ivan ask **who**₁ t₂?

This result is not compatible with my hypothesis above that the order of multiple specifiers is in principle free (though constrained by linearization). However, this tension is resolved by an additional constraint. Specifically, the correct result emerges from hypothesizing that tucking-in is only required for multiple movements to one head triggered by the same feature (McGinnis 1998, Rackowski 2002, Doggett 2004, Rackowski and Richards 2005, Kotek 2014):

- (21) [XP α_[F1] β_[F1] X_[uF1] ... t t] One attracting feature → tucking-in
- 

Thus I propose that in multiple *wh*-questions, all *wh*-phrases are attracted by one *wh*-sensitive feature on C, and a corresponding one on v, which force crossing-paths in vP and CP. In contrast, when the two phrases have different terminal positions and hence are ultimately attracted by separate features, I argue that v must bear the corresponding two features, which probe separately in either order, permitting crossing or nesting in vP. This optionality may be possible because nothing specifies which attracting feature must probe first:

(22) *Attraction by separate features on one head → Variability*

- a. [XP α_[F1] β_[F2] X_[uF1,uF2] ... t t] b. [XP β_[F2] α_[F1] X_[uF1,uF2] ... t t]
- 

That the probing features on v match the features of higher heads might be ensured by an agreement relation between the relevant heads, analogous to the theory in Rackowski and Richards (2005).

In summary, I propose that in multiple *wh*-questions where all *wh*-phrases have the same terminal position and the same type of semantic contribution, they are attracted by one feature, and hence must form crossing multiple specifiers.⁶ We find this in Bulgarian and English. In other scenarios where the relevant A'-moving phrases target separate landing sites, and hence are attracted by different features,⁷ we see flexibility of order in Bulgarian (18), though in English only nesting paths are allowed in such configurations for linearization reasons (9-13).⁸

⁶Of course, we also find superiority violating multiple-*wh* questions with discourse-linked phrases. The presence of multiple probes in such configurations may explain their unique properties: see Kotek (2014) for relevant arguments.

⁷In example (2a), both moved phrases land in the clausal periphery, potentially in separate positions (see footnote 1), but they clearly involve distinct movement types (*wh*-movement versus topicalization).

⁸A reviewer points out that the PCC also holds, for example, in *tough*-movement constructions where one of the moving elements is potentially a null operator:

5.1 Tucking-in with relation to the subject

In an English multiple-*wh* question where the first *wh*-phrase is an external argument subject, that subject undergoes overt movement, while the lower *wh*-phrase presumably undergoes covert movement. That covert movement should involve the formation of an inner specifier of vP below the origination position of the subject (23a). In contrast, in a single *wh*-question where the *wh*-phrase is a non-subject, that phrase moves across the external argument, presumably forming an outer specifier of vP above that subject's base generation position (23b):

- (23) a. Who₁ $\emptyset_2$ [_{vP} *t*₁ *t*₂ bought what₂]? b. What₂ did you₁ [_{vP} *t*₂ *t*₁ buy *t*₂]?

Thus it must be possible for successive-cyclic movement of a non-subject to form a specifier either above or below the external argument's base position in vP, as needed by linearization.

I have argued that freedom in specifier ordering is what we expect when two specifiers are formed by different features. This is the case in (23) as well. The external argument undergoes external merger in vP for independent reasons, and this process is perhaps itself feature-driven (Müller 2010). The internal merger implicated in successive-cyclic movement of the non-subject through vP is driven by a wholly separate feature. Since there are independent motivations for these two sorts of specifier formation, we expect their potential ordering to be free (though filtered out by linearization requirements). See Davis (2020b, ch. 5) for related discussion.

6 Conclusion

I have argued that facts about PGs indicate that in overlapping movement derivations, the final order of moved phrases matches their intermediate order in vP. This is as CL predicts. I further argued that the covertness of movement to an inner specifier in a language like English causes the PCC, and that the overt movement of inner specifiers in Bulgarian yields the lack of PCC in this language. I finally argued that tucking-in is only forced when one feature drives all movements, whereas attraction by separate features allows variability in specifier order, though linearization sometimes restricts the options that are licit at PF.⁹

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- (i) a. *[Which sonatas]₁ is this violin easy [_{CP} OP₂ to play *t*₁ on *t*₂]?
 b. [Which violin]₁ is the sonata easy [_{CP} OP₂ to play *t*₂ on *t*₁]? (Chomsky 1977, pg. 105)

As the reviewer remarks, the arising of the PCC in such cases suggests that null operators also participate in the determination of linear ordering. This is indeed argued for in Davis (2020b).

⁹This paper has focused on A-bar movement. See Davis (2020b, pg. 245, fn. 44) for an argument that these considerations also apply to A-movement.

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